Current Perspectives in Management Research

UU-DBA-820

Week 1

Topic Overview - The Role of Theory in Management Research

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Learning Outcomes

LO.1 Demonstrate an awareness of some of the issues currently being researched within management.

LO.2 Demonstrate an awareness of the methodological challenges associated with management research.

LO.5 Critique research, evaluate arguments and assess research presentations.

LO.12 Display the critical elements of a research plan relevant to the student's dissertation

Introduction

Theories are formulated to explain, predict, and understand phenomena and, in many cases, to challenge and extend existing knowledge within the limits of critical bounding assumptions. The theoretical framework is the structure that can hold or supp ort a theory of a research study. Theory and theorizing have long played an important role in the evolution and practice of science. Theory embodies statements of the knowledge that has been developed by humanity in a form that has both use in the practical world where human beings act based on their knowledge (partly learned from theories) and in the theoretical world where researchers validate or refute old knowledge and build new knowledge in the form of theories. As a collaborative practice of creating and validating knowledge, science and the scientists who practice it rely on the clear and succinct statements of theory, possibly/often together with attendant explanations of theory. Theory and theorising play key roles in both the natural and the social/behavioural sciences. The natural sciences have advanced primarily through the extensive development and testing of theory, largely through positivist methods. The social and behavioural sciences have also advanced through the development and testing of theory. However, the methods for social science have varied from positivist through interpretivist methods due to the complex and subjective nature of (most of) social reality. The subjective and complex nature of social reality and the interpretivist methods used make theory testing more difficult, but building and testing of theories are still important goals in social research. Both natural and social sciences are empirical in nature, with theory as the primary product (Venable, 2006).

The Theory Building Process

The building of theory occurs in two major stages – the descriptive stage and the normative stage. Within each of these stages, theory builders proceed through three steps. The the theory-building process iterates through these three steps again and again. In the past, management researchers have quite carelessly applied the term *theory* to research activities that pertain to only one of these steps. Terms such "utility theory" in economics, and "contingency theory" in organization design, for example, actually refer only to an individual step in the theory-building process in their respective fields. It is more useful to think of the term "theory" as a body of understanding that researchers build cumulatively as they work through each of the three steps in the descriptive and normative stages. In many ways, the term "theory" might better be framed as a verb, as much as it is a noun – because the body of understanding is continuously changing as scholars who follow this process work to improve it (Carlile & Christensen, 2005).

The Building of Descriptive Theory

The descriptive stage of theory building is a *preliminary* stage because researchers generally must pass through it in order to develop more advanced normative theory. The three steps that researchers who are building descriptive theory utilize are observation, categorization, and association.

Step 1: Observation

In the first step researchers observe phenomena and carefully describe and measure what they see. Careful observation, documentation and measurement of the phenomena in words and numbers is important at this stage because if subsequent researchers cannot agree upon the descriptions of phenomena, then improving theory will prove difficult. As per Argyris & Schon (1976), this stage of research is depicted in Figure 1 as the base of a pyramid because it is a necessary foundation for the work that follows. The phenomena being explored in this stage include not just things such as people, organizations and technologies, but processes as well. These observations can

be done anywhere along the continuum from analysis of huge databases on the one end, to field-based, ethnographic observation on the other.

Researchers in this step often develop what we term *constructs.* Constructs are abstractions that help us understand the essence of what the phenomena are and how they operate.

Step 2: Classification

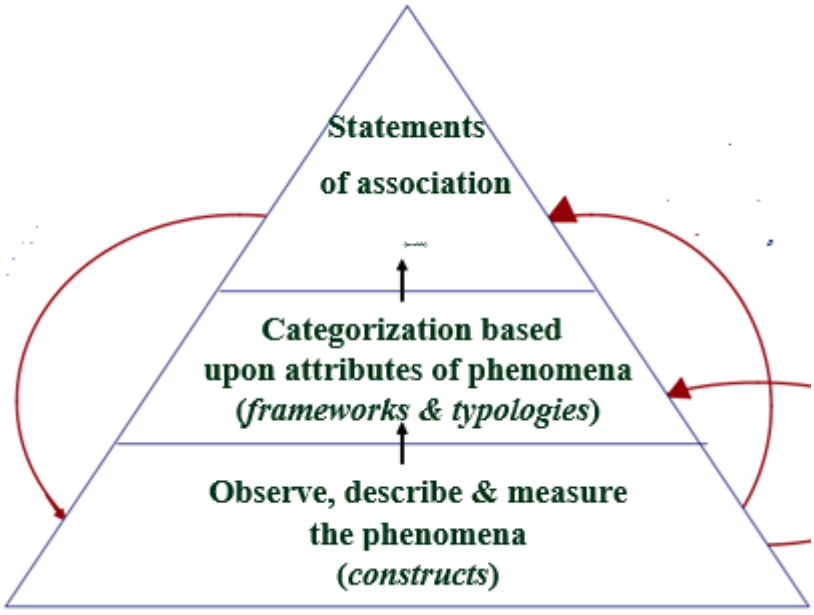
With the phenomena observed and described, researchers in the second stage then classify the phenomena into categories. In the descriptive stage of theory building, the classification schemes that scholars propose typically are defined by the attributes of the phenomena. Management researchers often refer to these descriptive categorization schemes as *frameworks* or *typologies*. Burgelman & Sayles (1986), for example, built upon Bower's (1970) construct of context by identifying two different types of context– organizational and strategic.

Step 3: Defining Relationships

In the third step, researchers explore the association between the category- defining attributes and the outcomes observed. In the stage of descriptive theory building, researchers recognize and make explicit what differences in attributes, and differences in the magnitude of those attributes, *correlate* most strongly with the patterns in the outcomes of interest. Techniques such as regression analysis typically are useful in defining these correlations. Often we refer to the output of studies at this step as *models*. Descriptive theory that quantifies the degree of correlation between the category-defining attributes of the phenomena and the outcomes of interest are generally able to make probabilistic statements of association representing average tendencies (Carlile & Christensen, 2005).

Figure 1

The Process of Building Theory by Carlile, P. R., & Christensen, C. M. (2005). *The cycles of theory building in management research*. Division of Research, Harvard Business School.



Descriptive theory-building efforts typically categorize by the attributes of the phenomena because attributes are easiest to observe and measure. Likewise, correlations between attributes and outcomes are easiest to hypothesize and quantify through techniques such as regression analysis. Kuhn (1962) observed that confusion and contradiction typically are the norm during descriptive theory-building. This phase is often characterized by a plethora of categorization schemes because the phenomena generally have many different attributes.

The confusion and contradiction that often accompany descriptive theory become resolved when careful researchers – often through detailed empirical and ethnographic observation – move beyond statements of correlation to define what *causes* the outcome of interest. Normative theory, like its descriptive predecessor, still needs to be improved – and researchers do this by following the same steps that were used in the descriptive stage. Hypothesizing that their statement of causality is correct, they cycle deductively to the bottom of the pyramid to test the causal statement: If we observe these actions being taken, these should be the outcomes that we observe.

When they encounter an anomaly, they then delve back into the lower levels of the pyramid. Sometimes they can resolve anomalies by developing more accurate, less ambiguous ways to define and measure the phenomena. Often they account for the anomalies by revisiting the categorization stage. Rather than using schemes based on attributes of the phenomena, however, in building normative theory researchers categorize *the different situations or circumstances* in which managers might find themselves. They do this by asking, when they encounter an anomaly, "What was it about the situation in which those managers found themselves, that caused the causal mechanism to yield a different result? By asking this question as they cycle up and down the pyramid of normative theory, anomaly-seeking researchers will ultimately define a relatively complete set of the situations or circumstances in which managers might find themselves when pursuing the outcomes of interest. This allows researchers to make contingent statements of causality – to show how and why the casual mechanism results in a different outcome, in the different situations. A normative theory that is built upon well-researched categories of circumstances can help a manager predict accurately what actions will and will not lead to the desired result, given the circumstance in which she finds herself. The relatively accurate, circumstance- contingent predictability of normative theory enables managers to know, in other words, what they *ought* to do (Carlile & Christensen, 2005).

Science, as a method, relies on both logic, as captured by theory, and empirical observation of the world to determine whether the theory we have developed conforms to what we actually observe. We seek to explain the world with our theories, and we test our theories by deducing and testing hypotheses (Argyris & Schon, 1976). Theory-testing with cases is ascertaining whether the empirical evidence in a case or in a sample of cases either supports or does not support the theory. There are two methodologies for theory testing with cases, (a) testing in a single case (theory-testing single case study), and (b) testing in a sample of cases (theory-testing sample case study). The functional form of the proposition that is tested determines which of these two methodologies should be used.

Applications: First the theory-testing single case study is discussed, and then the theory-testing sample case study (Hak & Dul, 2009)

Theory-Testing Single Case Study

Many theories have the form “X results in Y” or “X contributes to Y” or “X affects Y”, etc., in which X is, for example, something that an actor can or cannot do (or a situation or occasion or event that can occur or not occur) and Y is the desired outcome of that action. There are two fundamentally different ways of interpreting such a theoretical statement. Usually it is meant as a statement that explains differences between cases in the value of one (dependent) variable Y by relating them to differences in the values of another (independent) variable X: “If there is more X, then there is more Y”. Often, however, such a statement is meant to identify X as an important („critical‟ or „crucial‟) condition that should be present in order to make outcome Y possible. The intended meaning is that Y is very unlikely to occur if X is absent or, in other words, that Y normally is not possible without the presence of X. This is a necessary condition hypothesis (“Y only if X”). If necessary condition is present, this does not guarantee that the outcome will occur (which would imply that the condition is „sufficient‟) but only that the outcome has become possible. The concept of a necessary condition must not be confused with a sufficient condition. The necessary-condition proposition is an important type of theoretical statement. Necessary conditions and their outcomes can have different forms, but often they are discrete dichotomous (or dichotomized) variables (i.e., discrete categorical variables

with only two possible values, such as Present / Absent). The condition is a state, an event, or (in process theories) a sequence of events, that must be present in order to make the presence of the outcome possible. The implications of a necessary condition can be visualized as a cross-table as in Figure 1a, in which each dot represents a number of cases. The necessary condition is defined by the absence of cases in the cell “X absent/Y present” (Hak & Dul, 2009).

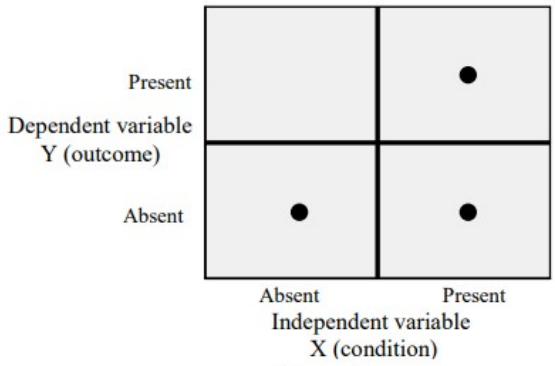


Figure 1a

Theoretical pattern of distribution of cases indicating a necessary condition.

Each dot represents a number of cases

Source: Dul & Hak (2008: 69)

Similarly, the implications of a sufficient condition (i.e., the statement that the outcome will be present if the condition is present) are visualized in Figure 1b. The sufficient condition is defined by the absence of cases in the cell “X present/Y absent”.

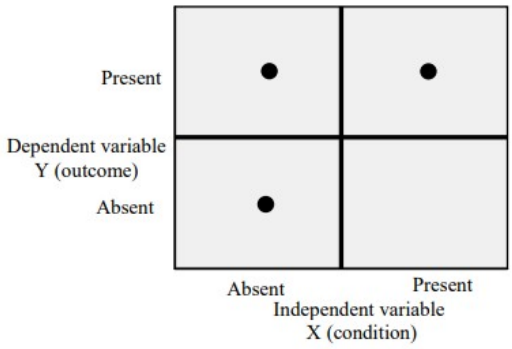


Figure 1b

Theoretical pattern of distribution of cases indicating a sufficient condition.

Each dot represents a number of cases

Source: Dul & Hak (2008: 68)

The methodology of the theory-testing single case study is quite simple and entails seven steps, which are very similar to the methodology of any other type of theory-testing study, be it a survey, an experiment or another type of research strategy. These steps are discussed here, using the example of the test of a necessary condition proposition (Hak & Dul, 2009).

Step 1. Formulate the theoretical statement that will be tested. The theoretical statement that is tested in a theory-testing single case study is a proposition about characteristics of single cases. A necessary condition statement is an example of such a proposition. The proposition “X is necessary for Y” is used here as an example.

Step 2. Select an appropriate case. One case of the theoretical domain is selected for the test. Criteria for case selection differ between the independent variable design and the dependent variable design (see the entry on Pattern matching). In the independent variable design, a case is selected in which Y is present (and an expected pattern is formulated about the value of the independent variable X). In the dependent variable design, a case is selected in which X is absent (and an expected pattern is formulated about the dependent variable Y).

Step 3. Specify the hypothesis for that case. The proposition must be translated into a hypothesis which can be formulated as an expected pattern. In the case in which the

outcome Y is present, the expected pattern is “X is present”. In the case in which the condition X is absent, the expected pattern is “Y is absent”.

Step 4. Measure the relevant variables. Criteria for valid and reliable measurement are the same for any type of research strategy, be it a survey, an experiment, a case study, or another theory-testing research strategy. In the present example, measurement entails ascertaining whether X is present or absent, or ascertaining whether Y is present or absent. The result of this measurement is the “observed pattern”.

Step 5. Test the hypothesis. Testing consists of comparing the observed pattern with the expected pattern. In the present example testing consists of determining whether X is present (in the independent variable design) or whether Y is absent (in the dependent variable design).

Step 6. Formulate the test result. The test result is either a disconfirmation of the hypothesis or a confirmation. In the present example, the hypothesis is disconfirmed if X is absent (in the independent variable design) or if Y is present (in the dependent variable design).

Step 7. Formulate the implications of the test result for the theory. Conclusions about the robustness of a theoretical statement cannot be drawn on the basis of just one test, but only after a series of tests. Hence, discussing the implications of a test result always implies comparing the result with those of earlier tests in a series of replications (see the entry on Replication). The procedure for testing a sufficient condition proposition follows the same logic. The differences are that other cases must be selected for the test and that, hence, other expected patterns are formulated. In the independent variable design, a case is selected in which Y is absent and the expected pattern is that X is absent as well. In the dependent variable design, a case is selected in which X is present and the expected pattern is that X is present as well.

Theory-Testing Sample Case Study

The methodology of the theory-testing sample case study is derived from how theoretical statements about differences between cases are tested in the sample

survey, i.e. in a study in which information is collected about a sample of cases from a population (usually by means of a standardized questionnaire) in order to draw conclusions about the population from which the sample was drawn. This entails the same seven steps as in the methodology of the theory-testing single case study. These steps are discussed, using the example of the test of a linear relationship (Hak & Dul, 2009).

Step 1. Formulate the theoretical statement that will be tested. The theoretical statement that is tested in a theory-testing sample case study is a proposition about differences between cases from the theoretical domain. A proposition that expresses a linear relationship, such as “If there is x% more X, then there is y% more Y”, is taken here as an example.

Step 2. Select an appropriate sample. As with a test in a sample survey, one population of cases from the theoretical domain must be selected for the test and a random sample is selected from that population.

Step 3. Specify the hypothesis for that sample. The proposition “If there is x% more X, then there is y% more Y” must be translated into an expected pattern. This translation (or specification) can have different forms, of which an expectation about the value of the regression slope bs in the sample probably is the most useful particularly if, in a replication study, a confidence interval of b is available from previous studies (Smithson, 2000).

Step 4. Measure the relevant variables. The scores of X and Y need to be measured validly and reliably in each case. This is not specific for this specific theory-testing strategy and, thus, the same procedures can be applied as in any other research strategy.

Step 5. Test the hypothesis. Pattern matching in the theory-testing sample case study consists of comparing an expected pattern (as specified in the hypothesis) with the observed pattern. When testing the hypothesis specified as an example in Step 3 above, it is determined whether the regression slope in the sample is in the specified range. Note that no attempt is made to confirm or disconfirm a null hypothesis

Step 6. Formulate the test result. A test result of a theory-testing sample case study is either a confirmation or a disconfirmation of the hypothesis (Smithson, 2005).

Step 7. Formulate the implications of the test result for the theory. As with the theory-testing single case study, the implications of the test result for the theory depend on the number of preceding tests as well as of the characteristics of the populations in which these other tests were conducted. Only after sufficient replications the research community might be able to draw a conclusion about the robustness of the theory.

References

Argyris, C., & Schon, D. (1976), *Theory in Practice*. San Francisco: Jossey-Bass.

Bower, Joseph (1970), *Managing the Resource Allocation Process*. Englewood Cliffs, NJ: Irwin.

Burgelman, R. & Sayles, L. (1986), *Inside Corporate Innovation*. New York: The Free Press.

Carlile, P. R., & Christensen, C. M. (2005). *The cycles of theory building in management research*. Division of Research, Harvard Business School.

Dul, J. & Hak, T. (2008). *Case Study Research Methodology in Business Research*. Oxford: Butterworth-Heinemann.

Dul, J., Hak, T., & Goertz, G. (2008). Testing necessary condition hypothesis with cases. Proceedings of the 15th international annual EUROMA conference, Groningen, the Netherlands, June 15-18, 2008.

Hak, T., & Dul, J. (2009). Theory-testing with cases.

Kuhn, T. (1962), *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press, 1962.

Smithson, M. (2000). Statistics with confidence. London: Sage.

Venable, J. (2006). The role of theory and theorising in design science research. In *Proceedings of the 1st International Conference on Design Science in Information Systems and Technology (DESRIST 2006)* (pp. 1-18).